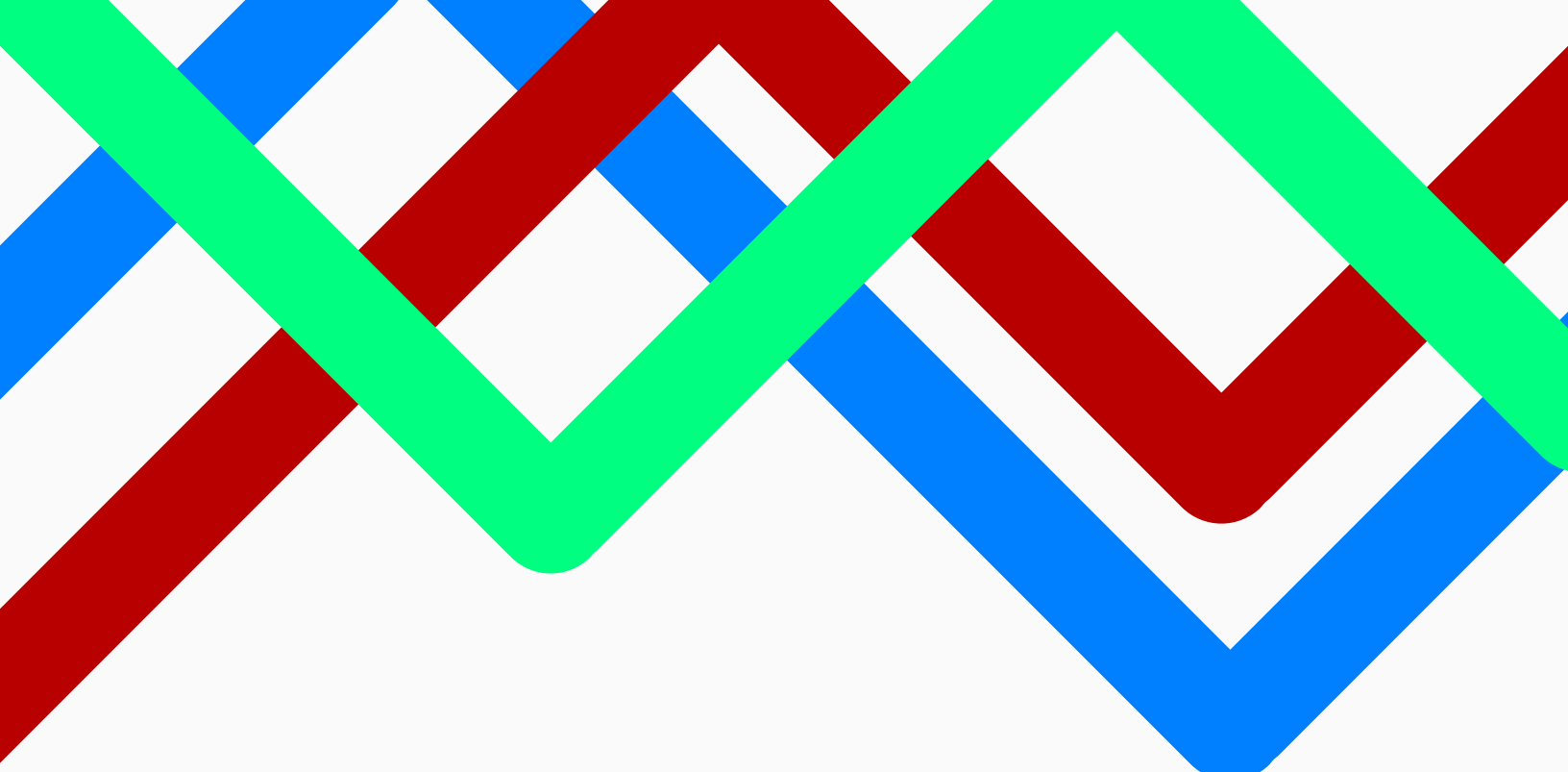




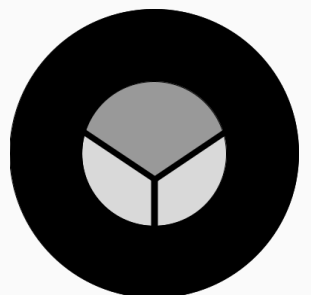
Unit 14 Review

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Trigonometry

October 7, 2025

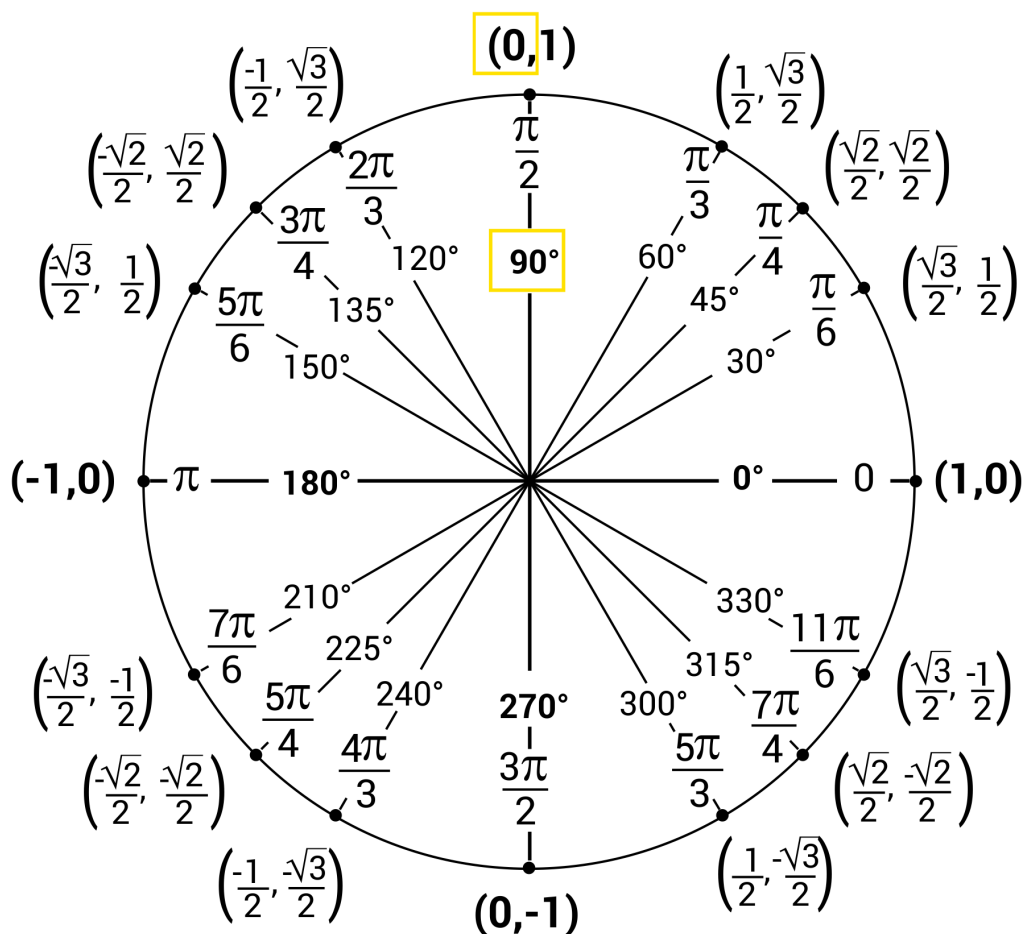
“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



1 Topics covered in quiz

- Graphing using Unit Circle
- Solving equations with general solutions / restrictive intervals
- Modeling sinusoidal functions given Maximum and Minimum

2 Unit Circle



3 Questions

3.1 Topic 1: Solving Equations with General Solutions / Restrictive Intervals

Solve the following equations. Assume the unknown variable is between $[0, \frac{\pi}{2})$ (In radians)

1. $2\cos(2x) - \sqrt{3} = 0$
2. $4\sin^2(\theta) - 4 = 1$
3. $3\tan(x) + \sqrt{3} = 0$
4. $\sec(u)\sqrt{3} = 2$
5. **Challenge:** $2\sin^2(z) * \cot^2(z) = 1$

Solve the following equations for their general solutions:

1. $\sin(x) = \frac{1}{2}$
2. $\cos(x) = -\frac{\sqrt{2}}{2}$
3. $\tan(x) = \sqrt{3}$
4. $\sin(x) = -\frac{\sqrt{3}}{2}$
5. **Challenge:** $2\cos^2(x) - 3\sin(x) = 0$

3.2 Topic 2: Modelling sinusoidal functions given Maximum and Minimums

Model the following three problems using Sine. Assume any point is either a maximum or a minimum point.

1. $(0, 0), (1, \pi)$
2. $(3, 2), (5, 4)$
3. $(\pi, \pi), (-\pi, -\pi)$

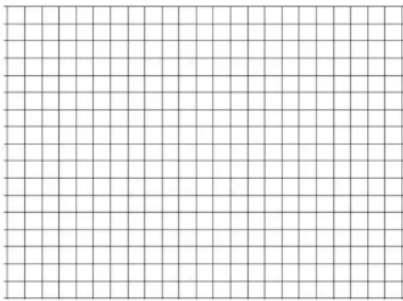
Model the following three problems using Cosine. Assume any point is either a maximum or a minimum point.

1. $(2, 2), (3, 4)$
2. $(0, 1), (2\pi, 3)$
3. $(2\pi, 2\pi), (-2\pi, -2\pi)$

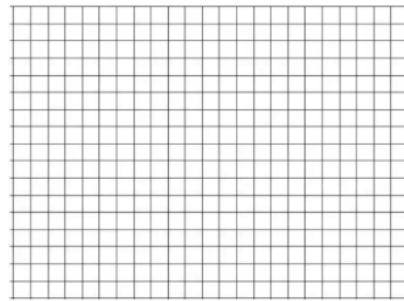
3.3 Topic 3: Graphing using Unit Circle

3.3.1 Graph the following problems

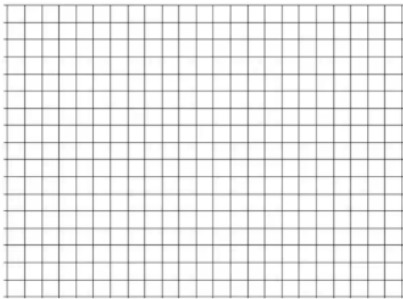
$$\sin(2x)$$



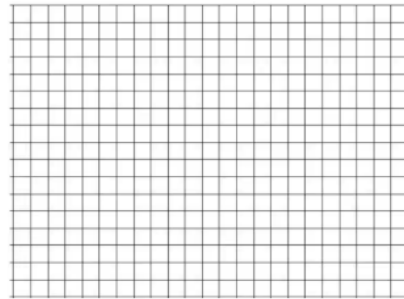
$$-3\cos(x)$$



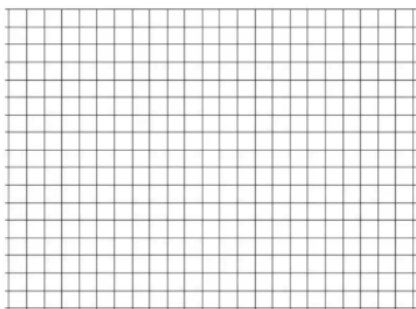
$$\cos(x-2\pi)$$



$$\sin(x)+5$$



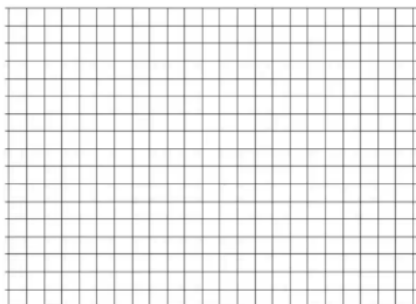
$$-\tan(x)$$



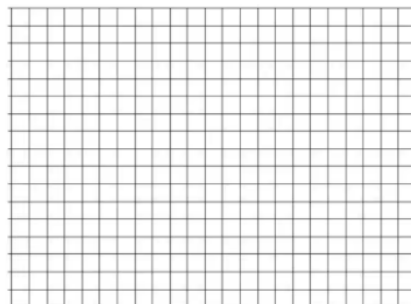
$$-3\cos(2x)$$



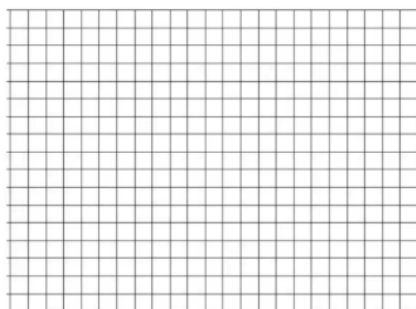
$$3\cos(x+2\pi)$$



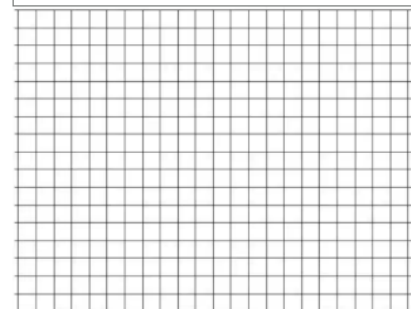
$$\tan(x)+2$$



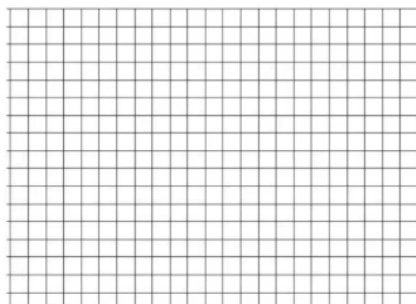
$$-\tan(3x)+5$$



$$-3\cos(2x-3\pi)$$



$$\tan(x-2\pi)$$



$$-2\tan(x)+2$$

